Team Project Check-In 1

Group Name: HeyBene
Group Members: Calerb Louis Jean, Haoran Huo, Xinqi Tao
Contact: cjean33@gatech.edu

1 INTRODUCTION

For our team project, we have chosen to redesign the user interface of the OpenBot (Müller and Koltun, 2021) robot car control application. OpenBot is an open-source project where a smartphone is mounted on a small self-assembled electric car, serving as both the computing hub and the camera. A second phone, or a gamepad, is then used to remotely control the car.

Our team has direct access to this hardware, and our starting point is an existing fork of the official app called OpenBene, which has already replaced the original app's backend with a more stable Bluetooth connection layer. Despite this technical foundation, the current interface has serious usability problems that make the control experience frustrating and unintuitive.

The most glaring issue is the phone-as-gamepad controller screen: the official design simply pastes a physical gamepad layout onto a touchscreen, ignoring entirely how people naturally hold and interact with a phone. Beyond that, the interface surfaces rarely-used technical parameters at all times, mode-switching between Drive, Auto Drive, and Tracking is unclear, and there is no live camera feed integrated into the control view.

Our goal is to redesign the control experience so that it feels as natural as a mobile racing game rather than a repurposed engineering debug screen (Delong et al., 2020).

2 NEEDFINDING PLAN

To understand the problem space from multiple angles, we will conduct user-centered needfinding activities targeting both experienced and first-time users, supplemented by an expert evaluation of the current interface against established usability heuristics.

2.1 Activity 1: Survey

For the first activity, we will run an online survey. A survey allows us to gather data from two distinct user types at once—people who have already used a robot control interface and people who have never encountered one—without requiring physical access to the hardware.

- **Participants & Recruitment:** We are targeting 15–25 participants drawn from two groups. The first group consists of existing OpenBot or robotics hobbyist users, recruited through direct outreach from our team's existing network (approximately 4–6 known contacts). The second group consists of new or first-time users with no prior robot-control experience, recruited from classmates, colleagues, and the course Ed discussion forum. We will post a brief recruitment message asking for five minutes of their time to help improve a robot control app for a class project.
- **Survey Design:** The survey is designed to surface real friction points without leading participants toward our own assumptions. We will avoid yes/no and agree/disagree questions entirely to prevent acquiescence bias. The survey is kept short to reduce satisficing, where respondents pick the first acceptable answer just to finish. We will use a mix of multiple-choice, Likert-scale, and open-ended questions so we capture both quantitative signal and rich qualitative context.
- **Data Goal:** We want to understand:
  1. what mental models users bring to phone-based robot control before they touch the app.
  2. which interaction patterns feel most natural on a touchscreen for directional input.
  3. and what information users actually need on screen while driving versus what can be hidden by default.

2.2 Activity 2: Semi-Structured Interviews

In addition to the survey, we plan to conduct 3–5 semi-structured interviews with participants from both target groups: people who have prior experience with robot control interfaces and people who do not. We believe interviews will help us go beyond preference selection and better understand why certain interaction patterns feel intuitive or confusing.

Each interview will last approximately 10–15 minutes and can be conducted remotely. During the interview, we will show participants screenshots or a brief demonstration of the current OpenBene/OpenBot control interface and ask them to reflect on their expectations, confusions, and preferred control styles.

Our interview goals are to better understand:

1. what users expect a phone-based robot control interface to look and feel like.
2. which elements of the current interface feel confusing, unnecessary, or overly technical.
3. what control patterns feel most natural on a touchscreen.
4. and what information users want to see during active driving.

Example interview questions include:

- When you first look at this control screen, what stands out to you?
- What seems confusing, unnecessary, or out of place?
- How would you expect to steer a robot car using a phone?
- What information would you want visible while driving?
- How would you expect switching between manual driving and automatic tracking to work?

2.3 Activity 3: Heuristic Evaluation

As a supplementary analysis to our user-centered needfinding activities, we will also conduct a heuristic evaluation of the current OpenBene control app. We will walk through three core tasks: connecting to the robot car and entering Drive mode; switching between Drive and Tracking mode; and using the phone-as-gamepad controller to steer the car. We will evaluate the interface against three of Nielsen's heuristics (Nielsen, 1994) that are most directly relevant to our redesign goals.

2.3.1 *Match Between System and the Real World*

The phone-as-gamepad screen is a direct violation of this heuristic: it maps a physical device metaphor (a gamepad with triggers, bumpers, and dual thumbsticks) onto a flat touchscreen that people hold completely differently. We will evaluate the extent to which the control vocabulary (button labels, icons, layout) matches how users actually think about steering a small car from their phone, rather than how an engineer thinks about a gamepad.

2.3.2 *Aesthetic and Minimalist Design*

The current interface exposes a dense parameter bar at all times, including values like motor speed coefficients and sensor thresholds that the vast majority of users never change during normal operation. Following Nielsen's principle that every extra unit of information competes with the relevant information, we will evaluate which UI elements earn persistent screen space and which should be collapsed into a settings panel accessed only when needed.

2.3.3 *Visibility of System Status*

When the car is connecting, driving, or switching modes, it is not always clear to the user what state the system is in. We will evaluate how well the app communicates connection status, active drive mode, and live telemetry—and specifically whether the absence of a live camera feed in the control view creates a critical gap in situational awareness that forces users to mentally model where the car is rather than seeing it directly.

3 FEEDBACK WE WANT

We would especially appreciate feedback on the following:

1. Whether our planned needfinding methods are appropriate for this stage of the project.
2. Whether our survey and interview plan will help us identify the current pain points in the OpenBot/OpenBene control experience.
3. Whether we are targeting the right participants, including both users with prior robot-control experience and first-time users.
4. Whether our questions will help us uncover the kinds of interaction styles users expect on a phone-based robot control interface.
5. Whether we are missing any important contexts, user groups, or usability concerns that should shape the redesign.

4 REFERENCES

1. Nielsen, Jakob (1994). "10 Usability Heuristics for User Interface Design". In: *Nielsen Norman Group*. URL: https://www.nngroup.com/articles/ten-usability-heuristics/.

2. Delong, M. L. et al. (2020). “Review of Human–Machine Interfaces for Small Unmanned Systems With Robotic Manipulators”. In: *IEEE Transactions on Human-Machine Systems* 50.3, pp. 250–264.
3. Müller, Matthias and Koltun, Vladlen (2021). “OpenBot: Turning Smartphones into Robots”. In: *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 5003–5009.

APPENDIX: SURVEY QUESTIONS

1. Have you ever used a remote-control car, drone, or robot before?
   - Yes — with a physical gamepad or remote
   - Yes — with a phone app
   - Yes — with both
   - No, never
2. If you have used a phone app to control a device before, how intuitive did it feel on your first try?
   - Scale 1–7 from "Extremely confusing" to "Extremely intuitive"; N/A option
3. Imagine you are holding your phone horizontally to steer a small robot car. Which input method would feel most natural to you?
   - Two on-screen thumbsticks like a gamepad
   - A single steering wheel you can slide left/right
   - Tilt the phone itself to steer
   - On-screen arrow buttons
   - Other (please describe below)
4. When driving a robot car from your phone, what information would you most want to see on screen at all times? (Select up to three)
   - Live camera feed from the car
   - Speed/motor output
   - Battery level
   - Current drive mode
   - Connection status
   - Nothing, keep the screen clean
5. How comfortable are you with technical settings like motor speed parameters or sensor thresholds appearing on your control screen during normal driving?
   - Scale 1–7 from "Very uncomfortable, they get in the way" to "Very comfortable, I want to see everything"
6. What, if anything, do you find most frustrating about controlling physical devices with your phone? (Open-ended)
7. Describe how you would expect to switch between "manual driving" and "automatic tracking" mode on a phone app. What would make that switch feel obvious and easy? (Open-ended)

Note: We will share the survey link with anyone interested in robot control or mobile UI, and post it in the Ed discussion forum to recruit participants.